
In-Parameter Learning: Why Lifelong AI Systems Need More Than Longer Context

Yiding Wang^{1,*}, Haotong Yang^{1,*}, Pingzhi Tang^{1,*}, Shijia Kang^{1,*},
Cai Zhou², Chenglin Zhu¹, Fan Jiang¹, Fanxu Meng¹,
Juntong Wang¹, Libin Chen¹, Tianyi Guan¹, Xiyuan Wang¹,
Xuejie Liu¹, Yanbo Wang¹, Yansheng Mao¹, Yewei Liu¹,
Yi Hu¹, Yufei Xu¹, Yuxuan Wang¹, Zian Li¹, Ziyuan He¹,
Xing Sun^{3,†}, Muhan Zhang^{1,†,‡}

¹Peking University

²Massachusetts Institute of Technology

³Tencent Youtu Lab, Shanghai, China

*Equal contribution. †Senior authors. Other authors are listed in alphabetical order.

‡Correspondence to: Muhan Zhang <muhan@pku.edu.cn>.

Abstract

AI agents are increasingly expected to interact with changing users, tools, and environments over extended periods of time. However, current systems remain largely *static*: their parameters are fixed during deployment, while new experiences, domain knowledge, corrections, and behavioral adaptations are handled primarily through external memory management and in-context learning. In this position paper, we focus on lifelong AI: AI systems that can continually learn from their long-term experience throughout an extended deployment lifetime. **This position paper argues that existing in-context mechanisms are insufficient for lifelong AI, and advocate In-Parameter Learning (IPL) as a necessary complement to In-Context Learning (ICL)**, referring to mechanisms that allow a deployed model to continually and safely update its parameters during use. More broadly, we aim to renew the community’s attention to in-parameter learning and to recalibrate the overly optimistic expectations placed on in-context learning alone. Our central claim is that future lifelong AI systems should be built around a hybrid paradigm: in-context learning for immediate, provisional, and reversible information, and in-parameter learning for durable, cumulative, and generalizable growth.

1 Introduction

The world an AI agent operates in is *non-static*: facts shift, tools and APIs change, domains drift, and users bring evolving goals, preferences, and working contexts. An agent deployed over extended periods must therefore be more than a capable problem-solver at a single moment, but an *active learner* that grows with its environment through interactions. We refer to this expectation as *Lifelong AI*: a system that continually accumulates experience and improves its capabilities throughout an extended deployment lifetime, rather than remaining fixed after release [Silver et al., 2013, Chen and Liu, 2018, Wang et al., 2024].

Today’s pretrained large language models (LLMs) are *static*: after a resource-intensive training process, models’ parameters are frozen, and all post-deployment adaptations are applied in the context window. Models rely on retrievers or search tools to obtain external knowledge during inference [Lewis et al., 2020, Izacard et al., 2023, Borgeaud et al., 2022, Nakano et al., 2021]; injected prompts reshape the model’s behavior [Ouyang et al., 2022, Sahoo et al., 2024], and context

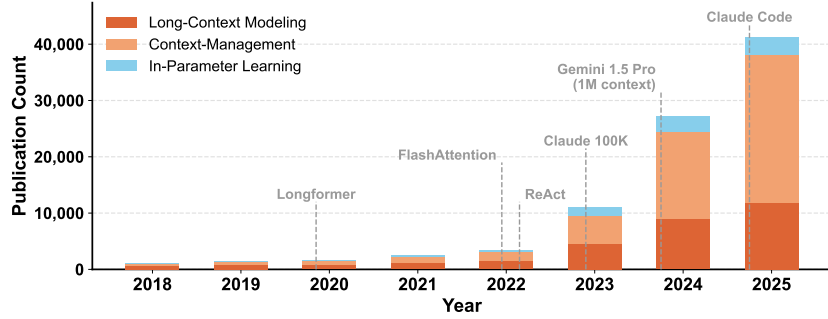


Figure 1: Publication trends from 2018 to 2025, based on annual OpenAlex [Priem et al., 2022] hit counts for three curated keyword queries. Long-context modeling and context management have grown rapidly into major LLM research directions, while continual In-Parameter Learning remains an emerging area.

management techniques [Zhong et al., 2024, Tan et al., 2025, Hu et al., 2025, Xu et al., 2025], accompanied by external documents, attempt to retain user history or project states across sessions; this has become the default practice in deployed agent systems [LangChain, 2025, Anthropic, 2026b, Manus, 2025]. The central problem with this approach is that its mode of control is inherently *unstable* and *ephemeral*. The context window is bounded by a hard limit, and anything beyond it is inaccessible to the model at generation time. As a result, once important instructions, factual knowledge, or historical interactions are pushed out of the available context, the model ceases to condition on them, and its behavior correspondingly drifts away from what is desired (i.e., *behavioral drift* [Rath, 2026]).

This is why so much recent work, in both research and practice, has focused on long context: either expanding context window, or managing context more carefully. In Figure 1, we gather the number of publications about LLM long context, showing the exponentially increasing trend. While these advances enable models to process significantly more information, inherent context limitations remain. Furthermore, scaling context length becomes progressively harder: continually extending the context window demands disproportionate advancements in architecture, computation, training data, and infrastructure [Nawrot et al., 2025, Gao et al., 2025, Chroma, 2025]. On the other hand, context management relies on approximate retrieval and lossy compression, constructing an unreliable external proxy that compounds errors over time and lacks the durable internal updates necessary for genuine knowledge accumulation and generalization [Jiang et al., 2024, Hosseini et al., 2022]. More fundamentally, all these In-Context Learning (ICL) paradigms leave the model’s parameters unchanged: no matter how well context is managed, the model’s intrinsic capabilities remain bounded by what its pretrained weights already encode.

As a result, we must return to the essence of LLMs themselves. Their vast parameter spaces encode enormous amounts of information, constituting their most central, durable and consequential store of knowledge and skills [Petrone et al., 2019]. Context, by contrast, is better viewed as an inference-time channel for temporal instructions, demonstrations, evidence, and transient environmental observations, enabling immediate adaptation without persistent parameter updates [Brown et al., 2020]. For lifelong AI, this distinction suggests that the parameter space should not remain a static artifact produced only before deployment, but should serve as the fundamental substrate through which useful knowledge, experience and skills are consolidated.

To this end, we emphasize that **In-Parameter Learning (IPL)**, the counterpart to ICL, deserves far greater attention and study. We use IPL here to refer to “*the timely and continual consolidation of newly acquired information into the model’s parameter space as enduring knowledge, behavioral habits, new skills and so on.*”¹

Such a capability promises more than simply overcoming the limits of finite context windows. Where ICL can interpret and use the supplied context only through the capabilities already encoded in the

¹We note that centralized paradigms such as pretraining and post-training also fall under the broad scope of IPL. However, our use of the term here is more specific: we focus on continual parameter updates during the model’s deployment, rather than centralized, large-scale training conducted before deployment.

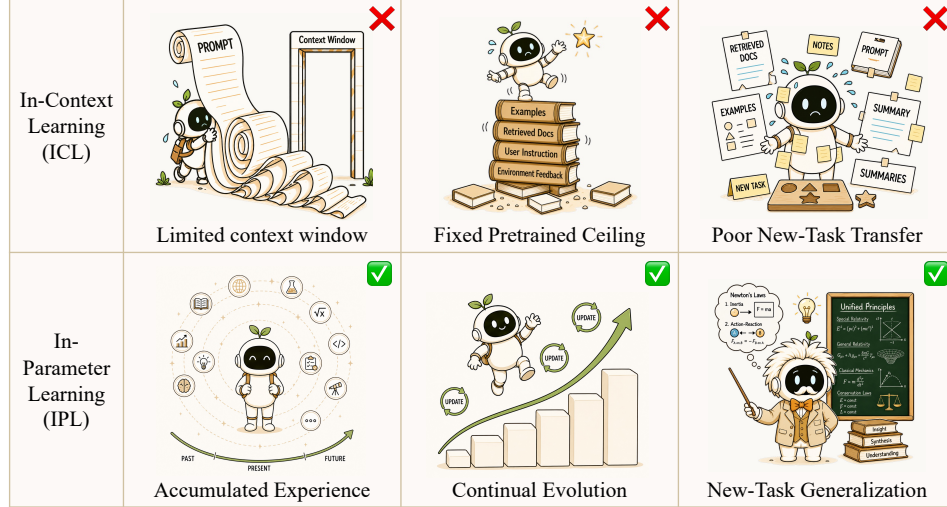


Figure 2: A comparison of ICL and IPL across three dimensions: the scale of experience each can accumulate, the capability ceiling each imposes, and the degree to which each supports novel-task generalization.

pretrained weights, IPL can raise this ceiling: by consolidating newly acquired information into parameters [Gururangan et al., 2020, Meng et al., 2022], the model can develop knowledge and skills previously inaccessible within the pretrained parameter space, generalize more robustly to out-of-distribution and compositionally novel tasks by reshaping its inductive biases [Mosbach et al., 2023, Lampinen et al., 2025], and reduce the recurring cost of re-processing the same evidence as unfamiliar, fragmented prompts at every inference step [Liu et al., 2024, Hsieh et al., 2024]. Figure 2 illustrates this contrast: ICL acts as the ordinary student who carries stacks of reference books into every exam, limited by what can fit in context, what the pretrained model can reliably leverage from them, and the growing overhead of processing them; IPL is the scholar who internalizes each lesson, compounding toward the kind of mastery no prompt can independently supply.

Importantly, this position should not be interpreted as a rejection of ICL. ICL retains crucial advantages (including transience, immediacy, and reversibility) and should work in tandem with IPL as the primary interface for external tools, user instructions, and short-term contextual adaptation. Therefore, the role of IPL is not to replace ICL’s function as the model’s working memory. Rather, it provides the vital pathway through which transient experience is elevated into permanent skills and raw knowledge is refined into competence. This role is analogous to long-term memory in the human brain: new knowledge is consolidated with lasting structural and synaptic changes rather than transient neural activity alone [Bailey et al., 1996, 2015]. Similarly, lifelong AI requires that experience be consolidated into parameters, rather than only held temporarily in context. **We argue that IPL should not be viewed as a dispensable extension to the current paradigm, but as a necessary path toward lifelong AI.**

To clarify and substantiate this position, the remainder of this paper is organized as follows.

- Section 2 argues that no foreseeable expansion of the context window can carry the experience a lifelong AI needs to accumulate, due to fundamental limits in scale, compute, and architecture.
- Section 3 argues that context management faces an inherent capability ceiling and cannot provide genuine self-evolution or internalization.
- Section 4 contrasts IPL with periodic large-scale retraining, illustrating how high-frequency parameter updates drive a self-evolving intelligence flywheel and enable parameter-level personalization.
- Section 5 discusses concrete technical directions toward realizing Lifelong AI systems in practice.

2 Why Extending Context Windows Cannot Suffice

This section argues that no foreseeable expansion of the context window can carry the information required by a lifelong AI. First, the volume of experience a lifelong AI must accumulate exceeds

the capacity of even today’s largest context windows by orders of magnitude (Section 2.1). Second, attempts to close this gap by further scaling context windows face fundamental obstacles in computation, data, and model architecture (Section 2.2). Finally, we argue that the parameter space of modern models, by contrast, offers a natural substrate to internalize what no context window can hold (Section 2.3).

2.1 Lifelong AI Requires More Context Than Any Window Can Hold

A lifelong AI must take in far more information than any context window can realistically accommodate. Consider language modality first: even a conservative estimate of 1.6×10^4 words per day [Mehl et al., 2007] amounts to roughly 4×10^8 tokens over a human lifetime, already **two orders of magnitude beyond today’s 1M-token frontier**. And this counts only spoken conversation; reading, writing, and inner speech [Hurlburt and Heavey, 2018] would push the estimate considerably higher, and for AI agents the inclusion of intermediate reasoning traces would widen the gap further [Guo et al., 2025].

The mismatch widens dramatically once **multi-modal and structured inputs** enter the picture. Raw sensory input is vastly larger still [Zheng and Meister, 2025], and video alone serves as a conservative lower bound: it captures only part of perceptual experience and is already heavily compressed by modern tokenizers, yet modern video tokenizers still require about 600K tokens per minute of video [Wan et al., 2025]. Scaled over years of experience, this reaches totals on the order of 10^{13} tokens. Structured inputs such as spreadsheets, relational databases, code repositories and tool-use traces [Sui et al., 2024, Fang et al., 2024, Wu et al., 2025] further compound the pressure, since their native structure must be serialized into long, often redundant token streams before an LLM can consume them [Dong et al., 2024]. Together, even conservative estimates place lifetime input across language, perception, and structured interaction at roughly **two to seven orders** of magnitude beyond current context capacity.

2.2 Scaling Context Windows Faces Multiple Bottlenecks

Although context windows have already grown rapidly, continuing to scale them runs into serious bottlenecks on multiple fronts. Frontier windows have expanded from GPT-3’s 2K tokens in 2020 to 1M+ tokens as the standard by early 2026, with specialized systems pushing 100M [Epoch AI, 2024b, Team et al., 2024, Magic, 2024]; we summarize the trajectory across major closed- and open-source models in Appendix A. The strain is already empirically visible: model performance often fails to scale proportionally with the supported context length [Kamradt, 2023, Bai et al., 2024, He et al., 2025b], the well-known “*lost-in-the-middle*” phenomenon causes information in the middle of long inputs to be ignored [Liu et al., 2024], and even perfect retrieval cannot prevent context length itself from hurting generation [Du et al., 2025]. In the following, we examine the structural sources of this strain along three axes: compute, data, and architecture.

Attention compute scales quadratically while practical resources do not. Self-attention requires $\Theta(n^2d)$ FLOPs per layer, and Keles et al. [2023] prove under SETH that no algorithm computes exact self-attention in strongly sub-quadratic time. KV cache memory is only $\Theta(n)$ per query, but it must be paid separately for each long-context request and cannot be amortized across a batch the way model weights can, so the per query memory footprint at lifelong-scale n rapidly outgrows the HBM of any single accelerator. Engineering optimizations such as FlashAttention [Dao et al., 2022], GQA/MQA [Ainslie et al., 2023], KV-cache quantization [Hooper et al., 2024], and sparse or sliding-window attention [Nawrot et al., 2025] reduce the constant factor or trade exactness for cost, but leave the dependence of $\Theta(n^2)$ compute and $\Theta(n)$ per-query memory untouched. Although aggregate frontier compute is growing rapidly [Epoch AI, 2024a], these resources are broadly distributed across training larger models, conducting additional runs, and supporting higher inference throughput, rather than being allocated for serving longer single-query contexts. Therefore, reaching the 10^9 – 10^{13} -token regime demanded by lifelong experience requires a fundamental regime change rather than a simple continuation of existing scaling trajectories.

Native long-context training data is scarce, and synthetic substitutes do not generalize. Training a natively long-context model requires high-quality long documents, which are in short supply. Long natural documents are dominated by code and books [Gao et al., 2025], and open-source instruction-

tuning corpora long enough to fill multi-hundred-thousand-token windows remain rare and small in scale [He et al., 2025a], motivating an entire line of synthetic long-context data engineering. Frontier models therefore typically avoid native long-context pretraining in favor of staged context extension. For example, Qwen3 pretrains at 32K and extrapolates to 128K during inference [Yang et al., 2025], while DeepSeek-V3 reaches 128K through two 1000-step YaRN extension stages on top of a 4K-pre-trained base [DeepSeek AI, 2024]. Although these resulting models successfully pass synthetic needle-in-a-haystack tests, their performance degrades on benchmarks that move past literal matching. In fact, real-world 32K inputs already noticeably degrade performance relative to short-context baselines [Modarressi et al., 2025].

Architectural mitigations reduce per-token cost but cannot remove the underlying floor. Sparse attention sparsifies which tokens the cache retains and history compression compresses each retained token; both lower per-token cost but neither removes the $O(n)$ memory floor. Sparse retention relies on heuristic importance scores that are fundamentally lossy and the achievable sparsity ratio is itself inherently bounded [Nawrot et al., 2025]; compression and quantization reduce per-token storage but introduce dequantization overhead, auxiliary metadata, and unpredictable quality degradation that compounds with context length [Yao et al., 2025].

2.3 Parameters as the Substrate for Accumulated Experience

The parameter spaces of modern LLMs, encompassing hundreds of billions to trillions of parameters, serve as a **natural repository for accumulated experience** that operates orthogonally to the context window. The interplay of linear transformations within the attention and MLP layers, combined with nonlinear activations, endows this space with an immense encoding capacity [Petroni et al., 2019] that far exceeds the limits of any finite context window. This architecture strongly parallels the consolidation of human memory: much like the hippocampus solidifies valuable, frequently accessed experiences while allowing irrelevant information to decay (a process reinforced during sleep and offline consolidation [Brodts et al., 2023, Cowan et al., 2021]), parametric updates allow models to **permanently internalize crucial knowledge**, providing a fundamental complement to context-based knowledge [Gururangan et al., 2020]. Knowledge encountered during deployment is absorbed into parameter updates, while context continues to supply information from the current interaction; together, they realize a **continuous spectrum of experience**, from the distant past encoded in parameters to the immediate present carried in context. This integration enables genuine access to lifelong experience rather than the bounded, fragmented approximation that context management alone can provide [Jiang et al., 2024, Du et al., 2025].

3 Why Context Management Cannot Suffice

Having shown that no amount of context expansion can accommodate lifelong experience, one might turn to context management: if the context window cannot grow indefinitely, perhaps sufficiently intelligent curation of its contents (through retrieval, compression, memory hierarchy, and agentic control) could approximate the effect of lifelong learning. We argue that this paradigm, while practically useful, fundamentally falls short of the requirements for lifelong AI. First, historically, virtually all step-changes in LLM capability have stemmed from parameter updates rather than advanced prompting (Section 3.1). Second, we demonstrate that in-context learning (ICL) is constrained by a strict capability ceiling dictated by the pretrained weights, a boundary that cannot be overcome simply by evolving the provided context (Section 3.2). Finally, we detail how IPL systematically breaks through this ceiling (Section 3.3).

3.1 The Driver of Progress Has Always Been Parameters

The history of large language models provides a consistent and informative lesson: **every major step-change in capability has been driven by improvements to the parameters** (through larger models, richer training data, better objectives, or more effective post-training techniques), not by improvements to prompting or context management. The revolution from GPT-2 to GPT-3 was not a consequence of a better prompt templates; it was the result of $100\times$ more parameters and vastly richer pretraining data [Brown et al., 2020]. Instruction-following emerged through RLHF, a form of parameter updating that reshaped the model’s internal representations rather than merely its input [Ouyang et al., 2022].

The recent wave of reasoning models makes this point especially sharp: although their capability surfaces through a chain-of-thought prompting format, the deliberative competence itself is acquired through large-scale reinforcement learning over reasoning trajectories [Guo et al., 2025], not conjured by the format alone.

Empirical evidence confirms that **tuning parameters outperform adjusting prompts on specific tasks**. When prompt tuning or related context adaptation methods are compared with full or parameter-efficient fine-tuning under matched conditions, parameter updates generally yield stronger performance, particularly on tasks that depart from the pretraining distribution [Lester et al., 2021]. Few-shot in-context learning, even on a substantial scale, falls behind parameter-efficient fine-tuning on many specialized tasks [Mosbach et al., 2023]. The interpretation is unambiguous: context can steer a model’s behavior within the envelope of what it has already learned but cannot extend that envelope. We should update the parameters to realize substantial model progress.

3.2 In-Context Learning Cannot Overcome Its Own Ceiling

Because ICL operates on fixed model parameters, the performance gains it yields on any given task are strictly bounded by what the model’s weights already encode. We demonstrate this fundamental ceiling from both theoretical and empirical perspectives.

ICL cannot surpass its pretrained foundations. The quality of in-context reasoning is bounded by the model’s intrinsic pretrained capacity for ICL. From a Bayesian perspective, ICL conditions the model on new observations without altering its prior; consequently, the posterior can only assign high probability to hypotheses already present within that prior [Xie et al., 2022]. Although in-context instructions or examples can successfully activate latent competencies, they cannot create new ones [Akyürek et al., 2023, von Oswald et al., 2023]. As a vivid illustration: even if a model were supplied with all existing literature on the Langlands Program within its context window, it could not produce a Fields-Medal-level proof, not for lack of information but for lack of the deeply internalized mathematical intuition that no prompt can independently supply.

ICL cannot improve ICL itself. Recent practices hope that iterative self-improvement, in which the model writes, refines, and accumulates its own skill files [Anthropic, 2025, Packer et al., 2024], could compound over time into expanded capability. This expectation has a fundamental limitation under an ICL-only paradigm. Reading a skill file is itself an act of ICL: the model’s parameters do not change, and therefore its capacity to interpret and apply the skills does not change either. Iterative writing can produce increasingly sophisticated artifacts, but the model’s ability to benefit from reading them is bounded by the same pretrained weights. As the gap between skill complexity and parametric interpretive capacity widens, self-evolved skills become either too complex to apply reliably or too abstract to be useful, and self-feedback accelerates this drift rather than correcting it. Zhong et al. [2026] confirm this empirically: no feedback-driven self-evolution approach matches skills written by human experts, and self-feedback alone causes performance to degrade across iterations.

Context management overhead compounds as during lifelong operation. As experience and skill documents accumulate, loading and reconciling an expanding set of external knowledge and guidance within a finite context window imposes an increasing burden. The underlying mechanisms are structurally unreliable: retrieval is approximate, since relevant evidence may be scattered across documents, left implicit in different conventions, or absent from any retrievable source [Lewis et al., 2020]; and any compression is lossy by design, with omitted reasoning steps and may distort omitted reasoning steps, implicit assumptions, or requirements across sessions [Jiang et al., 2024]. Even when retrieval succeeds, performance degrades as context length grows, because softmax attention dilutes the signal from relevant content as irrelevant tokens accumulate [Du et al., 2025]. Managing context thus becomes an ever-growing tax on capability rather than a path toward it: a self-defeating dynamic for any system meant to improve with use.

3.3 In-Parameter Learning Raises the Capability Ceiling

Reducing inference burden through knowledge internalization. In-Parameter Learning (IPL) allows a model to embed recurring knowledge and skills directly into its parameters, rather than carrying all relevant information in the context window for each inference. This reduces the cognitive

and computational load: the model no longer needs to “carry a large textbook” every time it solves a problem. In contrast, In-Context Learning (ICL) relies entirely on the context, requiring the model to process instructions or demonstrations repeatedly. IPL enables the model to approach tasks with information “already in mind”, analogous to a cook who has internalized fundamental cooking principles and can intuitively apply them without consulting a recipe for every dish [Silver et al., 2013, Chen and Liu, 2018]. This internalization accelerates reasoning and improves efficiency, especially as the experience grows.

Enhancing generalization through parametric learning. Large language models’ in-context learning exhibits distinct generalization behaviors compared to parameter-based learning. Multiple studies show that ICL can struggle with compositional generalization, where combinations of known primitives may fail to generalize to novel compositions, revealing a persistent compositional generalization gap in few-shot settings [Hosseini et al., 2022, Chan et al., 2022]. Mechanistically, these patterns reflect different inductive biases: generalization from contextual information is heavily exemplar-based, whereas transformers trained on parametric knowledge tend to generalize via parsimonious, internalized rules. This divergence aligns perfectly with the information-theoretic principle that intelligence emerges from data compression [Hutter, 2005, Delétang et al., 2024]. While ICL buffers raw, uncompressed information in a transient window, IPL forces the model to distill recurring experiences into incremental parameter updates. This parametric compression necessitates the discovery of deeper abstractions. By analogy, a cook who relies on ICL reads from a recipe book, able to prepare familiar dishes but fragile to new ingredient combinations, whereas an IPL-driven cook internalizes and compresses these experiences into fundamental culinary principles, allowing robust and systematic generalization over time [Mosbach et al., 2023, Lampinen et al., 2025].

IPL raises the model’s capability ceiling by internalizing novel knowledge and skills. Beyond improving efficiency and generalization, IPL enables models to acquire competencies fundamentally out of reach for static parameters [Akyürek et al., 2023, Zhong et al., 2026]. Because ICL operates over frozen weights, it acts essentially as an activation mechanism, limited to optimizing the use of a fixed, pre-existing capacity. In contrast, IPL incrementally assimilates continuous interactions, corrections, and preferences directly into the parameter space. This continuous consolidation reshapes the model’s inductive biases and intrinsic capabilities. By shifting from temporary contextual buffering to persistent weight modification, IPL transforms the model from a static reasoning engine into an evolving entity, serving as the critical engine for lifelong AI systems to grow continually.

4 From Periodic Retraining to Continual Evolution

The preceding sections establish that IPL is a necessary foundation for lifelong AI, which extends far beyond the limits of transient context management. A natural objection might accept this premise while defending the current industry standard: conducting large-scale periodic retraining, releasing updated model checkpoints, and treating each discrete cycle as a sufficient instance of IPL. We argue that while periodic retraining successfully advances baseline capabilities, **transitioning to continual IPL fundamentally alters the model’s evolutionary trajectory**. Rather than being bottlenecked by massive, human-curated engineering cycles, continual IPL operates at a high frequency, translating ongoing interactions into sensible, day-to-day capability enhancements. This rapid, automated capability improvement could kickstart a **self-evolving intelligence flywheel**. Furthermore, diverging from one-size-fits-all global checkpoints, such a continual and granular update mechanism naturally paves the way for parameter-level personalization, allowing models to dynamically tailor their weights to individual user experiences.

4.1 Periodic Retraining Incurs High and Redundant Costs

Each large-scale retraining cycle typically involves assembling extensive datasets, configuring complex training pipelines, and consuming weeks or months of high-performance compute resources. Even when only a modest amount of new information is added, these cycles often reprocess substantial portions of previously seen data to mitigate catastrophic forgetting [Kirkpatrick et al., 2017a]. Data curation, infrastructure coordination, and human annotation remain significant bottlenecks, meaning that capability improvements materialize only upon the cycle’s completion, often months after the initial experiences were acquired. Consequently, **the resource cost of retraining is disproportionate**

to the amount of new knowledge incorporated, making periodic retraining inherently less efficient for a system intended to grow continually.

In contrast, IPL is designed to operate incrementally. Parameter updates may be applied over small, targeted batches as a new experience arrives, without necessarily reprocessing the full historical corpus each time. This allows computational overhead to be proportional to the volume of new information rather than the full history of the model. While technical challenges remain (such as stability, avoidance of catastrophic forgetting, and update scheduling), incremental updates offer a plausible pathway to amortize learning cost over long deployment horizons.

4.2 Continual IPL Enables an Intelligence Flywheel and Personalization

Beyond computational cost, the arbitrary cadence of periodic retraining severely bottlenecks a model’s evolutionary trajectory. When capabilities are bound by human-managed, quarterly or annual release cycles, models suffer from a persistent adaptation lag. Critical feedback (such as user corrections, implicit preferences, and novel edge cases) remains trapped in an offline buffer until the next scheduled update. During this interval, errors persist, and the system remains static, preventing the model’s growth from matching the real-time velocity of its environment. Continual IPL fundamentally bypasses these human-specified bottlenecks, **transforming passive interactions into a self-sustaining “intelligence flywheel.”** By incrementally assimilating real-world learning signals directly into the parameter space, the model adapts autonomously as experiences occur. Crucially, this cumulative improvement accelerates itself: a richer parametric foundation also enables the model to absorb new contexts more efficiently and effectively. Rather than improving at a linear, human-constrained pace, the system’s capacity to learn might accelerate over time, allowing its competence to evolve increasingly faster as the flywheel gains momentum.

Furthermore, liberating the model from monolithic, globally synchronized updates **opens a direct pathway to parameter-level personalization.** Because continual IPL allows weights to update dynamically based on localized interaction streams, models can organically diverge from a rigid “one-size-fits-all” baseline. This high-frequency mechanism enables individual instances to deeply tailor their internal representations to specific user preferences, specialized workflows, and unique environmental contexts, laying the necessary groundwork for truly individualized lifelong AI.

5 Promising Directions Towards Realizing In-Parameter Learning

5.1 Parameter Updates Without Catastrophic Forgetting

Continuously integrating new knowledge into parameters without catastrophic forgetting remains a fundamental challenge. Classical continual learning methods like EWC [Kirkpatrick et al., 2017b] and GEM [Lopez-Paz and Ranzato, 2017] impose constraints from prior tasks, while parameter-efficient fine-tuning methods (e.g., LoRA) restrict update scope [Houlsby et al., 2019, Hu et al., 2022, Li and Liang, 2021]. However, these approaches are coarse-grained, externally applied, and assume clear task boundaries, limiting their applicability to open-ended lifelong AI [Wang et al., 2023, Jia et al., 2025]. Recent work on on-policy self-distillation further explores how newly acquired contextual knowledge can be internalized into parameters while mitigating catastrophic forgetting [Shenfeld et al., 2026]. Beyond such training-time mechanisms, a promising direction is native sparse update architectures, where selective parameter writing is an intrinsic property to the model design. Fully realizing this vision remains open, possibly requiring sophisticated gating mechanisms capable of dynamic write operations and structural plasticity to expand parameters without overwriting critical knowledge. Ultimately, future architectures should inherently balance stability and flexibility, enabling models to absorb novel experiences efficiently while preserving established capabilities.

5.2 Learning How to Update: Native Architectures and Meta-Learning

A deeper goal beyond designing better update mechanisms is to learn the update process itself, reducing reliance on hand-crafted retraining schedules and enabling more efficient and capable adaptation as deployment extends. Recent sub-quadratic architectures already perform a primitive form of in-parameter learning. State-space models [Gu and Dao, 2024] and the Titans/Atlas family [Behrouz et al., 2024, 2025] replace the growing KV cache with a fixed-size latent state that is updated by a learned online rule as each token arrives. From the IPL perspective, this state is itself a small set of

internal parameters, and the per-token update is like writing the context into these parameters, despite that the writable parameter space is much smaller than the entire model weight space. Building on this view, meta-learning, or “learning to learn”, provides a more general framework to transform context into parameters [Hospedales et al., 2021, Finn et al., 2017]. Hypernetworks can generate parameters conditioned on context, effectively mapping new experiences into the model’s weights [Ha et al., 2016, Munkhdalai and Yu, 2017, Chen et al., 2025a, Liang et al., 2025, Jukic et al., 2025, Charakorn et al., 2025, 2026, Liu et al., 2026]. Test-time training (TTT) offers a complementary approach: performing gradient updates during inference to compress newly observed data into parameters [Sun et al., 2025, Tandon et al., 2025]. Both approaches emphasize that knowledge should leave persistent traces in the model, not just in the ephemeral context.

5.3 Bridging Parameters and Context

Another practical challenge is building a system in which context and parameters work in concert: context handling what is immediate and transient, parameters encoding what is recurrent and durable, and a mechanism mediating the passage between them. The central challenge of this mediation is selective consolidation [Cowan et al., 2021, Brodt et al., 2023]: not all experience that passes through the context window deserve to be absorbed into parameters. A lifelong model must learn to distinguish signal from noise: privileging patterns that recur across sessions, corrections that reflect systematic error, and behavioral feedback that reshapes downstream performance, while filtering out low-quality, unreliable, or adversarially constructed inputs that should not be allowed to alter the model’s default computation. This is not merely a filtering problem but a deeper question of what it means for a model to learn from its own deployment. Naive absorption of all context information into parameters would be brittle and unsafe; indiscriminate consolidation risks encoding transient artifacts, conversational noise, or manipulative inputs as permanent behavioral tendencies.

5.4 Self-Evolving and Automatic IPL

A further promising direction is for the model to move beyond passive interaction: not merely updating from experience as it arrives, but actively identifying what it needs to learn, setting its own learning objectives, and generating or seeking out experience that targets its actual competence gaps. A passively updating system is at the mercy of whatever deployment happens to provide; a self-directing one monitors its own uncertainty, detects where its parametric knowledge is unreliable, and steers learning accordingly. Current self-training and self-play methods already show that models can improve themselves without human annotation [Zhao et al., 2025, Chen et al., 2025b, Liu et al., 2025], providing primitives for self-directed growth. Such self-directed IPL closes the loop between deployment and capability growth, removing the bottleneck of human curation and allowing improvement to scale with experience rather than with annotation effort. The ultimate expression of this is a model with an increasingly accurate model of its own competence, one that grows better at identifying its own gaps and more precise in targeting the experience that would address them.

5.5 Accelerating Scientific Discovery Through the Intelligence Flywheel

The technical directions outlined above lay the groundwork for a profound paradigm shift: realizing the “intelligence flywheel” introduced in Section 4.2. As continual IPL matures, the continuous internalization of new knowledge refines the model’s inductive biases, rendering subsequent learning cycles increasingly efficient and precise. This compounding effect holds transformative potential for AI-driven scientific discovery. Rather than functioning merely as an advanced literature retrieval engine, an IPL-enabled system acts as an evolving collaborator. It can organically assimilate novel experimental outcomes, adapt its internal representations to accumulating evidence, and generate increasingly sophisticated hypotheses based on deepened parametric intuition. Recent evidence supports this trajectory, demonstrating that targeted parametric updates empower models to surpass the performance of massive, parallel agentic search [Yuksekgonul et al., 2026]. This underscores a critical principle: in the pursuit of novel knowledge, the strategic internalization of learning signals fundamentally outweighs the sheer volume of context passively buffered. The intelligence flywheel, therefore, represents far more than a computational optimization; it signifies a qualitative leap in AI’s capacity to drive scientific discovery, serving as the ultimate motivation for the IPL research agenda.

6 Conclusion

ICL is fundamentally constrained by two distinct but related limitations: bounded scale and bounded capability. Overcoming these limits requires a paradigm shift towards In-Parameter Learning (IPL) as the essential substrate for continuous growth, supported by the concrete research directions outlined above. Context and parameters are not competitors but complements: context handles what is immediate and reversible, and parameters encode what is durable and generalizable. The path forward is co-evolution: both growing together, each doing what the other cannot. The research community has invested enormously in making the context longer and smarter. It is time to invest with equal seriousness in making the parameters live.

Lifelong AI will not be built on context alone.

References

- Joshua Ainslie, James Lee-Thorp, Michiel de Jong, Yury Zemlyanskiy, Federico Lebrón, and Sumit Sanghai. GQA: Training generalized multi-query transformer models from multi-head checkpoints. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 4895–4901, 2023.
- Ekin Akyürek, Dale Schuurmans, Jacob Andreas, Tengyu Ma, and Denny Zhou. What learning algorithm is in-context learning? investigations with linear models. In *ICLR*, 2023.
- Anthropic. Equipping agents for the real world with agent skills. <https://www.anthropic.com/engineering/equipping-agents-for-the-real-world-with-agent-skills>, 2025. Anthropic Engineering Blog, accessed 2026-04-20.
- Anthropic. Introducing claude opus 4.7. <https://www.anthropic.com/news/claude-opus-4-7>, 2026a.
- Anthropic. How claude remembers your project, 2026b. URL <https://code.claude.com/docs/en/memory>.
- Yushi Bai, Shangqing Tu, Jiajie Zhang, Hao Peng, Xiaozhi Wang, Xin Lv, Shulin Cao, Jiazheng Xu, Lei Hou, Yuxiao Dong, et al. Longbench v2: Towards deeper understanding and reasoning on realistic long-context multitasks. *arXiv preprint arXiv:2412.15204*, 2024.
- Craig H Bailey, Dusan Bartsch, and Eric R Kandel. Toward a molecular definition of long-term memory storage. *Proceedings of the National Academy of Sciences*, 93(24):13445–13452, 1996.
- Craig H Bailey, Eric R Kandel, and Kristen M Harris. Structural components of synaptic plasticity and memory consolidation. *Cold Spring Harbor perspectives in biology*, 7(7):a021758, 2015.
- Ali Behrouz, Peilin Zhong, and Vahab Mirrokni. Titans: Learning to memorize at test time. *arXiv preprint arXiv:2501.00663*, 2024.
- Ali Behrouz, Zeman Li, Praneeth Kacham, Majid Daliri, Yuan Deng, Peilin Zhong, Meisam Raza-viyayn, and Vahab Mirrokni. Atlas: Learning to optimally memorize the context at test time. *arXiv preprint arXiv:2505.23735*, 2025.
- Sebastian Borgeaud, Arthur Mensch, Jordan Hoffmann, Trevor Cai, Eliza Rutherford, Katie Millican, George Bm Van Den Driessche, Jean-Baptiste Lespiau, Bogdan Damoc, Aidan Clark, et al. Improving language models by retrieving from trillions of tokens. In *International conference on machine learning*, pages 2206–2240. PMLR, 2022.
- Svenja Brodt, Marion Inostroza, Niels Niethard, and Jan Born. Sleep—a brain-state serving systems memory consolidation. *Neuron*, 111(7):1050–1075, 2023. ISSN 0896-6273. doi: <https://doi.org/10.1016/j.neuron.2023.03.005>. URL <https://www.sciencedirect.com/science/article/pii/S0896627323002015>.
- Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901, 2020.

- Stephanie C. Y. Chan, Ishita Dasgupta, Junkyung Kim, Dharshan Kumaran, Andrew K. Lampinen, and Felix Hill. Transformers generalize differently from information stored in context vs in weights. *CoRR*, abs/2210.05675, 2022. doi: 10.48550/ARXIV.2210.05675. URL <https://doi.org/10.48550/arXiv.2210.05675>.
- Rujikorn Charakorn, Edoardo Cetin, Yujin Tang, and Robert Tjarko Lange. Text-to-lora: Instant transformer adaption. In *Forty-second International Conference on Machine Learning, ICML 2025, Vancouver, BC, Canada, July 13-19, 2025*. OpenReview.net, 2025. URL <https://openreview.net/forum?id=zWskCdu3QA>.
- Rujikorn Charakorn, Edoardo Cetin, Shinnosuke Uesaka, and Robert Tjarko Lange. Doc-to-lora: Learning to instantly internalize contexts. *CoRR*, abs/2602.15902, 2026. doi: 10.48550/ARXIV.2602.15902. URL <https://doi.org/10.48550/arXiv.2602.15902>.
- Tong Chen, Hao Fang, Patrick Xia, Xiaodong Liu, Benjamin Van Durme, Luke Zettlemoyer, Jianfeng Gao, and Hao Cheng. Generative adapter: Contextualizing language models in parameters with A single forward pass. In *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net, 2025a. URL <https://openreview.net/forum?id=bc3sUsS6ck>.
- Yixing Chen, Yiding Wang, Siqi Zhu, Haoifei Yu, Tao Feng, Muhan Zhang, Mostofa Patwary, and Jiaxuan You. Multi-agent evolve: Llm self-improve through co-evolution. *arXiv preprint arXiv:2510.23595*, 2025b.
- Zhiyuan Chen and Bing Liu. *Lifelong machine learning*. Morgan & Claypool Publishers, 2018.
- Chroma. Context rot. <https://www.trychroma.com/research/context-rot>, 2025.
- Emily T Cowan, Anna C Schapiro, Joseph E Dunsmoor, and Vishnu P Murty. Memory consolidation as an adaptive process. *Psychonomic Bulletin & Review*, 28(6):1796–1810, 2021.
- Tri Dao, Dan Fu, Stefano Ermon, Atri Rudra, and Christopher Ré. Flashattention: Fast and memory-efficient exact attention with io-awareness. *Advances in neural information processing systems*, 35: 16344–16359, 2022.
- DeepSeek AI. DeepSeek-V3 technical report. *arXiv preprint arXiv:2412.19437*, 2024.
- Grégoire Delétang, Anian Ruoss, Paul-Ambroise Duquenne, Elliot Catt, Tim Genewein, Christopher Mattern, Jordi Grau-Moya, Li Kevin Wenliang, Matthew Aitchison, Laurent Orseau, Marcus Hutter, and Joel Veness. Language modeling is compression. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2309.10668.
- Haoyu Dong, Jianbo Zhao, Yuzhang Tian, Junyu Xiong, Shiyu Xia, Mengyu Zhou, Yun Lin, José Cambronero, Yeye He, Shi Han, et al. Spreadsheetllm: Encoding spreadsheets for large language models. *arXiv preprint arXiv:2407.09025*, 2024.
- Yufeng Du, Minyang Tian, Srikanth Ronanki, Subendhu Rongali, Sravan Bodapati, Aram Galstyan, Azton Wells, Roy Schwartz, Eliu A Huerta, and Hao Peng. Context length alone hurts llm performance despite perfect retrieval. *arXiv preprint arXiv:2510.05381*, 2025.
- Epoch AI. Training compute of frontier AI models grows by 4–5× per year. <https://epoch.ai/blog/training-compute-of-frontier-ai-models-grows-by-4-5x-per-year>, 2024a.
- Epoch AI. Trends in context window sizes of frontier AI models. <https://epoch.ai/data-insights/context-windows/>, 2024b. Accessed: 2026-04-16.
- Xi Fang, Weijie Xu, Fiona Anting Tan, Jiani Zhang, Ziqing Hu, Yanjun Qi, Scott Nickleach, Diego Socolinsky, Srinivasan Sengamedu, and Christos Faloutsos. Large language models (llms) on tabular data: Prediction, generation, and understanding—a survey. *arXiv preprint arXiv:2402.17944*, 2024.
- Chelsea Finn, Pieter Abbeel, and Sergey Levine. Model-agnostic meta-learning for fast adaptation of deep networks. In *International conference on machine learning*, pages 1126–1135. PMLR, 2017.

- Chaochen Gao, Xing Wu, Zijia Lin, Debing Zhang, and Songlin Hu. Nextlong: Toward effective long-context training without long documents. *arXiv preprint arXiv:2501.12766*, 2025.
- Google DeepMind. Gemini 3.1 pro: A smarter model for your most complex tasks. <https://blog.google/innovation-and-ai/models-and-research/gemini-models/gemini-3-1-pro/>, 2026.
- Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. In *Conference on Language Modeling (COLM)*, 2024. arXiv:2312.00752.
- Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, Qihao Zhu, Runxin Xu, Ruoyu Zhang, Shirong Ma, Xiao Bi, et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- Suchin Gururangan, Ana Marasović, Swabha Swayamdipta, Kyle Lo, Iz Beltagy, Doug Downey, and Noah A Smith. Don’t stop pretraining: Adapt language models to domains and tasks. In *Proceedings of the 58th annual meeting of the association for computational linguistics*, pages 8342–8360, 2020.
- David Ha, Andrew Dai, and Quoc V Le. Hypernetworks. *arXiv preprint arXiv:1609.09106*, 2016.
- Linda He, Jue Wang, Maurice Weber, Shang Zhu, Ben Athiwaratkun, and Ce Zhang. Scaling instruction-tuned llms to million-token contexts via hierarchical synthetic data generation. *arXiv preprint arXiv:2504.12637*, 2025a.
- Ziyuan He, Yuxuan Wang, Jiaqi Li, Kexin Liang, and Muhan Zhang. Loogole v2: Are llms ready for real world long dependency challenges? *arXiv preprint arXiv:2510.22548*, 2025b.
- Coleman Hooper, Sehoon Kim, Hiva Mohammadzadeh, Michael W Mahoney, Yakun S Shao, Kurt Keutzer, and Amir Gholami. Kvquant: Towards 10 million context length llm inference with kv cache quantization. *Advances in Neural Information Processing Systems*, 37:1270–1303, 2024.
- Timothy Hospedales, Antreas Antoniou, Paul Micaelli, and Amos Storkey. Meta-learning in neural networks: A survey. *IEEE transactions on pattern analysis and machine intelligence*, 44(9): 5149–5169, 2021.
- Arian Hosseini, Ankit Vani, Dzmitry Bahdanau, Alessandro Sordani, and Aaron C. Courville. On the compositional generalization gap of in-context learning. In Jasmin Bastings, Yonatan Belinkov, Yanai Elazar, Dieuwke Hupkes, Naomi Saphra, and Sarah Wiegrefe, editors, *Proceedings of the Fifth BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP, BlackboxNLP@EMNLP 2022, Abu Dhabi, United Arab Emirates (Hybrid), December 8, 2022*, pages 272–280. Association for Computational Linguistics, 2022. doi: 10.18653/V1/2022.BLACKBOXNLP-1.22. URL <https://doi.org/10.18653/v1/2022.blackboxnlp-1.22>.
- Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin De Laroussilhe, Andrea Gesmundo, Maziar Attariyan, and Sylvain Gelly. Parameter-efficient transfer learning for NLP. In *Proceedings of the 36th International Conference on Machine Learning*, pages 2790–2799, 2019. URL <https://proceedings.mlr.press/v97/houlsby19a.html>.
- Cheng-Ping Hsieh, Simeng Sun, Samuel Krizan, Shantanu Acharya, Dima Rekesch, Fei Jia, and Boris Ginsburg. RULER: What’s the real context size of your long-context language models? In *Conference on Language Modeling (COLM)*, 2024.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022. URL <https://openreview.net/forum?id=nZeVKeeFYf9>.
- Mengkang Hu, Tianxing Chen, Qiguang Chen, Yao Mu, Wenqi Shao, and Ping Luo. Hiagent: Hierarchical working memory management for solving long-horizon agent tasks with large language model. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 32779–32798, 2025.
- Russell T Hurlburt and Christopher L Heavey. Inner speaking as pristine inner experience. 2018.

- Marcus Hutter. *Universal artificial intelligence: Sequential decisions based on algorithmic probability*, volume 300. Springer, 2005.
- Gautier Izacard, Patrick Lewis, Maria Lomeli, Lucas Hosseini, Fabio Petroni, Timo Schick, Jane Dwivedi-Yu, Armand Joulin, Sebastian Riedel, and Edouard Grave. Atlas: Few-shot learning with retrieval augmented language models. *Journal of Machine Learning Research*, 24(251):1–43, 2023.
- Ziqi Jia, Anmin Wang, Xiaoyang Qu, Xiaowen Yang, and Jianzong Wang. Hierarchical-task-aware multi-modal mixture of incremental LoRA experts for embodied continual learning. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 28415–28427, 2025. doi: 10.18653/v1/2025.acl-long.1379. URL <https://aclanthology.org/2025.acl-long.1379/>.
- Huiqiang Jiang, Qianhui Wu, Xufang Luo, Dongsheng Li, Chin-Yew Lin, Yuqing Yang, and Lili Qiu. Longllmlingua: Accelerating and enhancing llms in long context scenarios via prompt compression, 2024. URL <https://arxiv.org/abs/2310.06839>.
- Josip Jukic, Martin Tutek, and Jan Snajder. Context parametrization with compositional adapters. *CoRR*, abs/2509.22158, 2025. doi: 10.48550/ARXIV.2509.22158. URL <https://doi.org/10.48550/arXiv.2509.22158>.
- Greg Kamradt. Needle in a haystack - pressure testing llms. https://github.com/gkamradt/LLMTest_NeedleInAHaystack, 2023. Accessed: 2025-05-01.
- Feyza Duman Keles, Pruthuvi Mahesakya Wijewardena, and Chinmay Hegde. On the computational complexity of self-attention. In *Proceedings of the 34th International Conference on Algorithmic Learning Theory (ALT)*, 2023. arXiv:2209.04881.
- James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A. Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, Demis Hassabis, Claudia Clopath, Dharshan Kumaran, and Raia Hadsell. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13):3521–3526, March 2017a. ISSN 1091-6490. doi: 10.1073/pnas.1611835114. URL <http://dx.doi.org/10.1073/pnas.1611835114>.
- James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A. Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, et al. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13):3521–3526, 2017b. doi: 10.1073/pnas.1611835114.
- Andrew K. Lampinen, Arslan Chaudhry, Stephanie C. Y. Chan, Cody Wild, Diane Wan, Alex Ku, Jörg Bornschein, Razvan Pascanu, Murray Shanahan, and James L. McClelland. On the generalization of language models from in-context learning and finetuning: a controlled study. *CoRR*, abs/2505.00661, 2025. doi: 10.48550/ARXIV.2505.00661. URL <https://doi.org/10.48550/arXiv.2505.00661>.
- LangChain. Context engineering for agents, 2025. URL <https://www.langchain.com/blog/context-engineering-for-agents>.
- Brian Lester, Rami Al-Rfou, and Noah Constant. The power of scale for parameter-efficient prompt tuning. In *Conference on Empirical Methods in Natural Language Processing*, 2021. URL <https://api.semanticscholar.org/CorpusID:233296808>.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in neural information processing systems*, 33: 9459–9474, 2020.
- Xiang Lisa Li and Percy Liang. Prefix-tuning: Optimizing continuous prompts for generation. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 4582–4597, 2021. doi: 10.18653/v1/2021.acl-long.353. URL <https://aclanthology.org/2021.acl-long.353/>.

- Zhiyuan Liang, Dongwen Tang, Yuhao Zhou, Xuanlei Zhao, Mingjia Shi, Wangbo Zhao, Zekai Li, Peihao Wang, Konstantin Schürholt, Damian Borth, Michael M. Bronstein, Yang You, Zhangyang Wang, and Kai Wang. Drag-and-drop llms: Zero-shot prompt-to-weights. *CoRR*, abs/2506.16406, 2025. doi: 10.48550/ARXIV.2506.16406. URL <https://doi.org/10.48550/arXiv.2506.16406>.
- Bo Liu, Chuanyang Jin, Seungone Kim, Weizhe Yuan, Wenting Zhao, Ilya Kulikov, Xian Li, Sainbayar Sukhbaatar, Jack Lanchantin, and Jason Weston. Spice: Self-play in corpus environments improves reasoning. *arXiv preprint arXiv:2510.24684*, 2025.
- Nelson F Liu, Kevin Lin, John Hewitt, Ashwin Paranajpe, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the association for computational linguistics*, 12:157–173, 2024.
- Yewei Liu, Xiyuan Wang, Yansheng Mao, Yoav Gelbery, Haggai Maron, and Muhan Zhang. SHINE: A scalable in-context hypernetwork for mapping context to lora in a single pass. *CoRR*, abs/2602.06358, 2026. doi: 10.48550/ARXIV.2602.06358. URL <https://doi.org/10.48550/arXiv.2602.06358>.
- David Lopez-Paz and Marc’Aurelio Ranzato. Gradient episodic memory for continual learning. *Advances in neural information processing systems*, 30, 2017.
- Magic. 100M token context windows. <https://magic.dev/blog/100m-token-context-windows>, 2024.
- Manus. Context engineering for ai agents: Lessons from building manus, 2025. URL <https://manus.im/blog/Context-Engineering-for-AI-Agents-Lessons-from-Building-Manus>. By Yichao “Peak” Ji.
- Matthias R. Mehl, Simine Vazire, Nairán Ramírez-Esparza, Richard B. Slatcher, and James W. Pennebaker. Are women really more talkative than men? *Science*, 317(5834):82, 2007.
- Kevin Meng, Arnab Sen Sharma, Alex Andonian, Yonatan Belinkov, and David Bau. Mass-editing memory in a transformer. *arXiv preprint arXiv:2210.07229*, 2022.
- Mistral AI. Mistral small 4. <https://mistral.ai/news/mistral-small-4>, 2026.
- Ali Modarressi, Hanieh Deilamsalehy, Franck Dernoncourt, Trung Bui, Ryan A Rossi, Seunghyun Yoon, and Hinrich Schütze. Nolima: Long-context evaluation beyond literal matching. *arXiv preprint arXiv:2502.05167*, 2025.
- Marius Mosbach, Tiago Pimentel, Shauli Ravfogel, Dietrich Klakow, and Yanai Elazar. Few-shot fine-tuning vs. in-context learning: A fair comparison and evaluation. In Anna Rogers, Jordan L. Boyd-Graber, and Naoaki Okazaki, editors, *Findings of the Association for Computational Linguistics: ACL 2023, Toronto, Canada, July 9-14, 2023*, Findings of ACL, pages 12284–12314. Association for Computational Linguistics, 2023. doi: 10.18653/V1/2023.FINDINGS-ACL.779. URL <https://doi.org/10.18653/v1/2023.findings-acl.779>.
- Tsendsuren Munkhdalai and Hong Yu. Meta networks. In Doina Precup and Yee Whye Teh, editors, *Proceedings of the 34th International Conference on Machine Learning, ICML 2017, Sydney, NSW, Australia, 6-11 August 2017*, Proceedings of Machine Learning Research, pages 2554–2563. PMLR, 2017. URL <http://proceedings.mlr.press/v70/munkhdalai17a.html>.
- Reiichiro Nakano, Jacob Hilton, Suchir Balaji, Jeff Wu, Long Ouyang, Christina Kim, Christopher Hesse, Shantanu Jain, Vineet Kosaraju, William Saunders, et al. Webgpt: Browser-assisted question-answering with human feedback. *arXiv preprint arXiv:2112.09332*, 2021.
- Piotr Nawrot, Robert Li, Renjie Huang, Sebastian Ruder, Kelly Marchisio, and Edoardo M Ponti. The sparse frontier: Sparse attention trade-offs in transformer llms, 2025. URL <https://arxiv.org/abs/2504.17768>, 2025.
- OpenAI. Introducing GPT-5.4. <https://openai.com/index/introducing-gpt-5-4/>, 2026.

- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *Advances in neural information processing systems*, 35:27730–27744, 2022.
- Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. Memgpt: Towards llms as operating systems, 2024. URL <https://arxiv.org/abs/2310.08560>.
- Fabio Petroni, Tim Rocktäschel, Sebastian Riedel, Patrick Lewis, Anton Bakhtin, Yuxiang Wu, and Alexander Miller. Language models as knowledge bases? In *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP)*, pages 2463–2473, 2019.
- Jason Priem, Heather Piwowar, and Richard Orr. Openalex: A fully-open index of scholarly works, authors, venues, institutions, and concepts. *arXiv preprint arXiv:2205.01833*, 2022.
- Qwen Team. Qwen3.5: Multimodal hybrid reasoning LLMs. <https://github.com/QwenLM/Qwen3.5>, 2025.
- Abhishek Rath. Agent drift: Quantifying behavioral degradation in multi-agent llm systems over extended interactions. *arXiv preprint arXiv:2601.04170*, 2026.
- Pranab Sahoo, Ayush Kumar Singh, Sriparna Saha, Vinija Jain, Samrat Mondal, and Aman Chadha. A systematic survey of prompt engineering in large language models: Techniques and applications. *arXiv preprint arXiv:2402.07927*, 1, 2024.
- Idan Shenfeld, Mehul Damani, Jonas Hübner, and Pulkit Agrawal. Self-distillation enables continual learning. *arXiv preprint arXiv:2601.19897*, 2026.
- Daniel L Silver, Qiang Yang, and Lianghao Li. Lifelong machine learning systems: Beyond learning algorithms. In *AAAI Spring Symposium: Lifelong Machine Learning*, volume 13, 2013.
- Yuan Sui, Mengyu Zhou, Mingjie Zhou, Shi Han, and Dongmei Zhang. Table meets llm: Can large language models understand structured table data? a benchmark and empirical study. In *Proceedings of the 17th ACM International Conference on Web Search and Data Mining*, pages 645–654, 2024.
- Yu Sun, Xinhao Li, Karan Dalal, Jiarui Xu, Arjun Vikram, Genghan Zhang, Yann Dubois, Xinlei Chen, Xiaolong Wang, Sanmi Koyejo, Tatsunori Hashimoto, and Carlos Guestrin. Learning to (learn at test time): Rnns with expressive hidden states. In Aarti Singh, Maryam Fazel, Daniel Hsu, Simon Lacoste-Julien, Felix Berkenkamp, Tegan Maharaj, Kiri Wagstaff, and Jerry Zhu, editors, *Forty-second International Conference on Machine Learning, ICML 2025, Vancouver, BC, Canada, July 13-19, 2025*, Proceedings of Machine Learning Research. PMLR / OpenReview.net, 2025. URL <https://proceedings.mlr.press/v267/sun25h.html>.
- Zhen Tan, Jun Yan, I-Hung Hsu, Rujun Han, Zifeng Wang, Long Le, Yiwen Song, Yanfei Chen, Hamid Palangi, George Lee, et al. In prospect and retrospect: Reflective memory management for long-term personalized dialogue agents. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8416–8439, 2025.
- Arnub Tandon, Karan Dalal, Xinhao Li, Daniel Kocaja, Marcel Rød, Sam Buchanan, Xiaolong Wang, Jure Leskovec, Sanmi Koyejo, Tatsunori Hashimoto, Carlos Guestrin, Jed McCaleb, Yejin Choi, and Yu Sun. End-to-end test-time training for long context. *CoRR*, abs/2512.23675, 2025. doi: 10.48550/ARXIV.2512.23675. URL <https://doi.org/10.48550/arXiv.2512.23675>.
- Gemini Team et al. Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context. *arXiv preprint arXiv:2403.05530*, 2024.
- Johannes von Oswald, Eyvind Niklasson, Ettore Randazzo, João Sacramento, Alexander Mordvintsev, Andrey Zhmoginov, and Max Vladymyrov. Transformers learn in-context by gradient descent. In *ICML*, 2023.

- Team Wan, Ang Wang, Baole Ai, Bin Wen, Chaojie Mao, Chen-Wei Xie, Di Chen, Feiwu Yu, Haiming Zhao, Jianxiao Yang, et al. Wan: Open and advanced large-scale video generative models. *arXiv preprint arXiv:2503.20314*, 2025.
- Liyuan Wang, Xingxing Zhang, Hang Su, and Jun Zhu. A comprehensive survey of continual learning: Theory, method and application. *IEEE transactions on pattern analysis and machine intelligence*, 46(8):5362–5383, 2024.
- Xiao Wang, Tianze Chen, Qiming Ge, Han Xia, Rong Bao, Rui Zheng, Qi Zhang, Tao Gui, and Xuanjing Huang. Orthogonal subspace learning for language model continual learning. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 10701–10716, 2023. doi: 10.18653/v1/2023.findings-emnlp.715. URL <https://aclanthology.org/2023.findings-emnlp.715/>.
- Xiaofeng Wu, Alan Ritter, and Wei Xu. Tabular data understanding with llms: A survey of recent advances and challenges. *arXiv preprint arXiv:2508.00217*, 2025.
- xAI. Grok 3 beta — the age of reasoning agents. <https://x.ai/news/grok-3>, 2025.
- Sang Michael Xie, Aditi Raghunathan, Percy Liang, and Tengyu Ma. An explanation of in-context learning as implicit Bayesian inference. In *International Conference on Learning Representations (ICLR)*, 2022.
- Wujiang Xu, Yujie Liang, et al. A-Mem: Agentic memory for LLM agents. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.
- Dingyu Yao, Chenxu Yang, Zhengyang Tong, Zheng Lin, Wei Liu, Jian Luan, and Weiping Wang. Vecinfer: Efficient llm inference with low-bit kv cache via outlier-suppressed vector quantization. *arXiv preprint arXiv:2510.06175*, 2025.
- Mert Yuksekogul, Daniel Kocaja, Xinhao Li, Federico Bianchi, Jed McCaleb, Xiaolong Wang, Jan Kautz, Yejin Choi, James Zou, Carlos Guestrin, et al. Learning to discover at test time. *arXiv preprint arXiv:2601.16175*, 2026.
- Andrew Zhao, Yiran Wu, Yang Yue, Tong Wu, Quentin Xu, Matthieu Lin, Shenzhi Wang, Qingyun Wu, Zilong Zheng, and Gao Huang. Absolute zero: Reinforced self-play reasoning with zero data. *arXiv preprint arXiv:2505.03335*, 2025.
- Jieyu Zheng and Markus Meister. The unbearable slowness of being: Why do we live at 10 bits/s? *Neuron*, 113(2):192–204, 2025.
- Shanshan Zhong, Yi Lu, Jingjie Ning, Yibing Wan, Lihan Feng, Yuyi Ao, Leonardo F. R. Ribeiro, Markus Dreyer, Sean Ammirati, and Chenyan Xiong. Skilllearnbench: Benchmarking continual learning methods for agent skill generation on real-world tasks, 2026. URL <https://arxiv.org/abs/2604.20087>.
- Wanjun Zhong, Lianghong Guo, Qiqi Gao, He Ye, and Yanlin Wang. Memorybank: Enhancing large language models with long-term memory. In *Proceedings of the AAAI conference on artificial intelligence*, volume 38, pages 19724–19731, 2024.

A Context Window Growth Across Frontier Models

The past few years have seen an extraordinary expansion of context windows. Frontier models have grown from GPT-3’s 2K-token window in 2020 to over 1M tokens as the standard for every major frontier model by early 2026, at roughly $30\times$ per year since mid-2023 [Epoch AI, 2024b]. Google’s Gemini 1.5 Pro first broke the 1M barrier in February 2024 [Team et al., 2024]; by 2026 this scale has been matched by Gemini 3.1 Pro [Google DeepMind, 2026], Anthropic’s Claude Opus 4.7 [Anthropic, 2026a], OpenAI’s GPT-5.4 [OpenAI, 2026], and xAI’s Grok 4.1 [xAI, 2025], while specialized systems such as Magic’s LTM-2-mini push to 100M tokens [Magic, 2024]. The open-source ecosystem has converged on 128K–1M as the baseline, with Qwen3.5 natively supporting 256K (extensible to 1M) [Qwen Team, 2025], DeepSeek-V3.2 at 128K [DeepSeek AI, 2024], and Mistral Small 4 at 256K [Mistral AI, 2026].